

A SEVEN-VERTEX COUNTEREXAMPLE TO CONJECTURE 5.3 OF GÖBEL AND MISRA

ABSTRACT. Göbel and Misra conjectured that every equality between determinant polynomials of colored paths is generated by identity, reflection, or one of two parity-dependent constructions. We refute the odd-order clause of their Conjecture 5.3. Two uniformly vertex-colored paths on seven vertices, with edge words

$$bbcbcc \quad \text{and} \quad bccbcb,$$

have the same determinant polynomial but satisfy none of the conjectured alternatives. An exact exhaustive computation shows that seven vertices are minimal. The even-order clause remains open.

Let a colored path on m vertices have vertex word $u = u_1 \cdots u_m$ and edge word $w = w_1 \cdots w_{m-1}$, with vertex and edge colors drawn from disjoint alphabets. Associate variables x_c to vertex colors and y_d to edge colors, and write

$$K(u, w) = \begin{pmatrix} x_{u_1} & y_{w_1} & & & \\ y_{w_1} & x_{u_2} & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & & y_{w_{m-1}} & \\ & & & & x_{u_m} \end{pmatrix}, \quad D(u, w) = \det K(u, w).$$

If M is a matching of the path and $V(M)$ is the set of vertices incident with M , then

$$(1) \quad D(u, w) = \sum_M (-1)^{|M|} \prod_{i \in M} y_{w_i}^2 \prod_{j \notin V(M)} x_{u_j}.$$

This is the matching expansion in [1, Lemma 3.3].

Conjecture 5.3(2) of [1] asserts that, for odd m , equality of two such determinant polynomials arises only from identity, reflection, or the coloring configuration of [1, Theorem 3.8], up to reflection. In reading that theorem, we replace the edge-index ranges in its condition (3) by $\{1, \dots, m-1\}$, as required for a path on m vertices and as used by the matrices in its proof. In particular, its edge condition makes each parity class of edge positions monochromatic and exchanges the two colors between the paths.

Theorem 1. *Let a be a vertex color and let $b \neq c$ be edge colors. Put*

$$p = bbcbcc, \quad q = bccbcb, \quad x = x_a, \quad B = y_b^2, \quad C = y_c^2.$$

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Then

(2)

$$D(a^7, p) = D(a^7, q) = x^7 - 3x^5(B+C) + 2x^3(B^2 + 3BC + C^2) - 2xBC(B+C).$$

The two colored paths satisfy none of the alternatives in Conjecture 5.3(2). Hence that clause, and therefore Conjecture 5.3 as stated, is false.

Proof. For a six-letter edge word z , define

$$A_r(z) = \sum_{\substack{M \text{ a matching} \\ |M|=r}} \prod_{i \in M} y_{z_i}^2.$$

Since all seven vertices have the same color, (1) gives

$$D(a^7, z) = x^7 - x^5 A_1(z) + x^3 A_2(z) - x A_3(z).$$

The size-two matchings are indexed by

$$13, 14, 15, 16, 24, 25, 26, 35, 36, 46,$$

and the size-three matchings by

$$135, 136, 146, 246.$$

Substitution of either p or q therefore gives

$$\begin{aligned} A_1(p) &= A_1(q) = 3B + 3C, \\ A_2(p) &= A_2(q) = 2B^2 + 6BC + 2C^2, \\ A_3(p) &= A_3(q) = 2B^2C + 2BC^2. \end{aligned}$$

This proves (2).

The edge words are distinct, and the reverse of p is $c b c b b$, not q . Their odd- and even-position edge subsequences are

	odd positions	even positions
p	(b, c, c)	(b, b, c)
q	(b, c, b)	(c, b, c)

None is monochromatic. Reversal only reverses and exchanges these two subsequences, so no choice of orientations, and no interchange of the two paths, satisfies the edge condition of Theorem 3.8. Thus none of the conjectured alternatives applies. $\square$

Remark 2. For every real specialization with $x > 2 \max\{|y_b|, |y_c|\}$, both matrices in Theorem 1 are positive definite by strict diagonal dominance. Thus the equality occurs inside the positive-definite parameter space of the corresponding colored Gaussian graphical models.

MINIMALITY

For $n \geq 1$, let $\mathcal{W}_n$ be the set of all surjective words on $\{1, \dots, r\}$, for some $1 \leq r \leq n$, whose color multiplicities are weakly decreasing. When multiplicities tie, all label orders are retained.

Proposition 3. *No counterexample to Conjecture 5.3 has at most six vertices, under the orientation-invariant reading of the two constructions used above. Consequently, the counterexample in Theorem 1 is smallest with respect to the number of vertices.*

Proof. Equality of determinant polynomials forces equality of the vertex-color multiplicities, because the edge-degree-zero term in (1) is $\prod_j x_{u_j}$. It also forces equality of the edge-color multiplicities: after setting every vertex variable equal to 1, the coefficient of the pure monomial y_d^2 is minus the number of edges of color d . A common relabeling of the vertex colors and, independently, of the edge colors therefore sends every candidate pair to $\mathcal{W}_m \times \mathcal{W}_{m-1}$.

For each $(u, w) \in \mathcal{W}_m \times \mathcal{W}_{m-1}$ with $3 \leq m \leq 6$, we evaluated (1) as an exact integer coefficient dictionary indexed by the full exponent vector and grouped paths with identical dictionaries. Every unordered pair of distinct paths in a group was first tested for reflection. Each remaining pair was then tested against Theorem 3.6 when m is even and the type-correct conditions of Theorem 3.8 when m is odd, allowing the two paths to be interchanged and each path to be independently reversed. The disjoint counts are

m	paths	equal pairs	reflections	constructions	residual
3	30	14	14	0	0
4	470	248	232	16	0
5	11,562	5,780	5,772	8	0
6	394,092	197,496	197,016	480	0

The cases $m \leq 2$ are immediate from $D(u_1, \emptyset) = x_{u_1}$ and $D(u_1 u_2, w_1) = x_{u_1} x_{u_2} - y_{w_1}^2$.

□

The counterexample refutes precisely the odd-order clause of Conjecture 5.3. The exhaustive computation verifies the even-order clause only through six vertices; the general even-order clause remains open.

REFERENCES

- [1] H. Göbel and P. Misra, *Linear relations of colored Gaussian cycles*, arXiv:2506.23936v1 [math.CO], 2025.